

Acer Nitro 5 (AN515-55) Troubleshooting Report

1. Computer Identification:

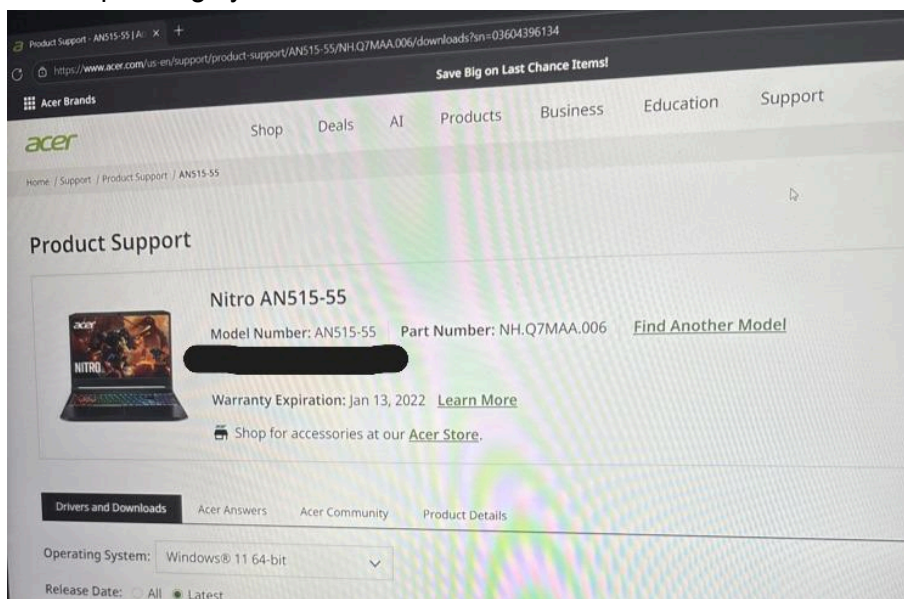
- Make: Acer
- Model: Nitro 5
- Exact Model Number: AN515-55
- Part Number: NH.Q7MAA.006
- Type: Gaming Laptop
- CPU: Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz (10th Generation Intel Core processor)
- RAM: 16384 MB (16GB)
- Operating System (Current): Windows 11 64-bit (upgraded from Windows 10)
- Storage: WDC PC SN530 SDBPNPZ-2566-1014 (NVMe SSD)
- Warranty Expiration: January 13, 2022

2. Initial Problem Description (Prior to July 26, 2025):

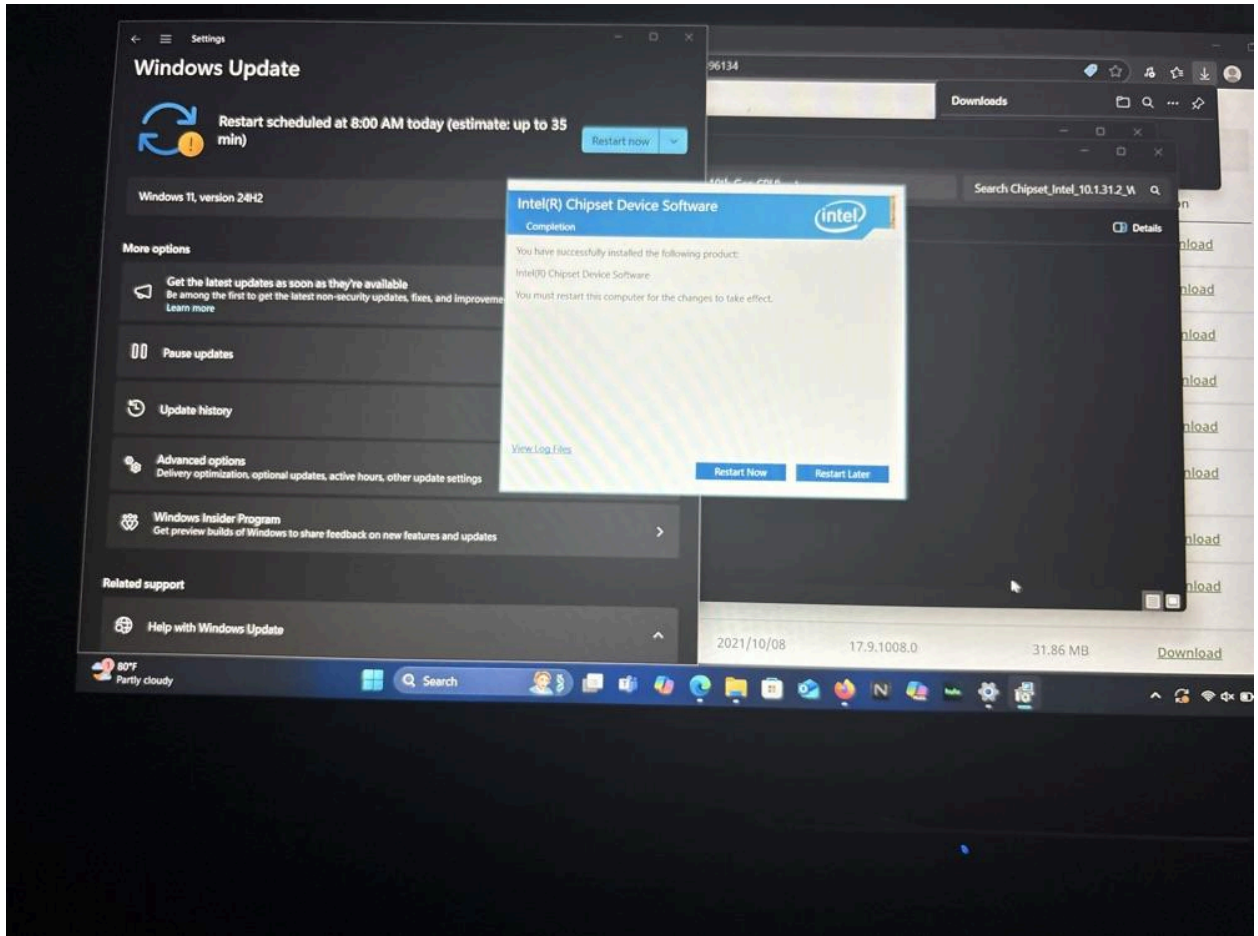
After upgrading the operating system from Windows 10 to Windows 11, the laptop exhibited an unacceptably slow boot time. Initially, the boot process was approximately 1 minute and 30 seconds (90 seconds) from power button press to desktop. A significant portion of this delay, specifically around 60 seconds, occurred before the Acer logo even appeared on the screen.

3. Visual Evidence (from provided images and their relevance to the report):

- Image 1 (Acer Product Support Page - Model Info):. This image confirms the laptop's model, part number, and warranty expiration date, as well as the selected Windows 11 64-bit operating system for driver downloads.



- Image 2 (Windows Update Screen): This image shows "Windows 11, version 24H2" in the downloading phase and "Acer Incorporated - HIDClass - 1.0.0.12" as completed.



- Image 4 (Downloaded Chipset Driver Folder): This image shows the contents of the unzipped "Chipset_Intel_10.1.31.2_W11x64_(for 10th Gen CPU)" folder, containing the SetupChipset.exe installer.
- Image 5 (Intel Chipset Software Installer Prompt): This image shows the "Intel(R) Chipset Device Software" setup completed and prompting for a restart, overlapping with Windows Update settings showing 24H2 restart pending.

4. Symptoms Observed and Chronological Events (July 26, 2025):

- Initial Boot Time (Prior to Chipset Update): Approximately 1 minute 30 seconds (90 seconds) total boot time, with 60 seconds for the Acer logo to appear.
- Chipset Driver Update (July 26, 2025):
 - Action: Navigated to Acer's official support website for the AN515-55 model. Downloaded the latest available "Chipset Driver (for 10th Gen CPU)" (Version 10.1.31.2, dated 2021/10/08).
 - Installation: Extracted the .zip file and ran SetupChipset.exe.
 - Restart Priority: User was advised to prioritize the chipset driver restart over the concurrently pending Windows 11 version 24H2 update.
 - Initial Post-Chipset Boot Result: User reported it was "the fastest I ever seen it load up!" However, precise timing was not yet recorded.

- Cold Boot Re-test after Chipset Driver (July 26, 2025):
 - Action: A full shutdown and subsequent power-on was performed for accurate cold boot timings.
 - Result: The Acer logo appeared in 47 seconds, and the total boot time to desktop was 1 minute and 35 seconds (95 seconds). This showed some improvement from the original 60-second pre-logo delay, but it was still significantly slow.
- BIOS/UEFI "Fast Boot" Setting Adjustment (July 26, 2025):
 - Action: Accessed BIOS (pressing F2 at boot). Confirmed current BIOS version was V2.06.
 - Discovery: The "Fast Boot" option was found under the "Main" tab after using a specific key combination (Ctrl + S) to reveal hidden options. It was initially set to [Enabled].
 - Adjustment: Changed "Fast Boot" from [Enabled] to [Disabled].
 - Re-test Result: The logo appeared in 42 seconds (a slight improvement from 47 seconds), and the total boot time was 1 minute 25 seconds (85 seconds).
- Windows 11 Version 24H2 Update (Completion - July 26, 2025):
 - Action: User prioritized completing the Windows 11 version 24H2 update, including its required restarts, to avoid conflicts with potential further BIOS adjustments.
 - Final Post-24H2 Re-test Result: After the 24H2 update was fully installed and the system settled, a full shutdown and re-timing was performed. The Acer logo appeared in 47 seconds, and the total boot time was 95 seconds (1 minute 35 seconds). This indicates that the update did not resolve the core boot delay.

5. Diagnosis and Remaining Possibilities for Slow Boot:

- Despite extensive troubleshooting involving critical drivers (chipset), major OS updates (24H2), and BIOS settings (Fast Boot), a significant delay persists in the Power-On Self-Test (POST) phase before the Acer logo even appears.
- Current BIOS/UEFI Firmware Limitation: Since the laptop is running the latest available BIOS (V2.06 from Acer's website for the AN515-55), it's possible that this specific BIOS version inherently has a relatively long POST sequence. Not all laptop models are equally optimized for extremely fast boot times, and some have more thorough or time-consuming hardware initialization routines built into their firmware.
- Subtle Hardware Initialization Delay: There might be a specific internal component that takes an unusually long time for the BIOS to detect or initialize during POST. This isn't necessarily a "failure" of the component, but a delay in its readiness signal to the BIOS. Diagnosing such a specific component without specialized tools is very difficult.
- Deep-seated Software/OS Corruption (Less Likely for Pre-Logo Delay): While less probable to cause the pre-logo delay, an extremely persistent software conflict or OS

corruption from the Windows 10 to 11 upgrade could contribute to overall boot time. The only way to definitively rule this out is a clean installation of Windows 11.

6. Recommended Future Actions:

- Given the expired warranty and the current situation-
 - Monitor Performance: Continue to use the laptop as normal. Observe if the boot time worsens over time, or if any other performance issues or system instability arise once Windows has fully loaded.
 - Consider a Clean Installation of Windows 11 (Optional Last Resort for Software): If the persistent boot time remains unacceptable, a full, clean reinstallation of Windows 11 (wiping the drive and installing from scratch) is the ultimate software troubleshooting step. This requires backing up all data and reinstalling all applications. This might improve the overall boot time, but its impact on the pre-logo delay is uncertain.
 - Contact Acer Support (for Information Only): Even though the warranty is expired, you can still contact Acer support. Explain all the steps you've taken and the precise boot timings. Ask if the ~47-second pre-logo POST time is considered normal for the AN515-55 model or if they have any internal diagnostic insights for this specific issue. Do not expect free repair, but they might offer valuable information.